Yucheng (Cecilia) Zhang

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Education

Northwestern University, M.S. Computer Science — Sep. 2026 – May 2028, Evanston, IL
The University of Texas at Austin, B.S. Mathematics; CS Minor; GPA: 3.74/4.00 — Aug. 2023 – May 2026, Austin, TX
Relevant Coursework: Data Structures, Database Systems, Data Analytics, Probability and Statistics, Stochastic Processes, Artificial Intelligence

Technical Skills

Languages: Python, SQL, R, Java, JavaScript, TypeScript, MATLAB, SAS
Data Engineering: ETL pipelines, data ingestion, data transformation, data validation, reconciliation, feature pipelines, reporting automation
Databases & Cloud: SQL Server, PostgreSQL, Oracle, Azure Synapse, Databricks, Delta Lake, AWS, Docker
Tools & Analytics: Pandas, NumPy, Power BI, Git, REST API development, SQLAlchemy, Alembic, Excel, Scikit-learn

Experience

Enterprise Risk Analyst Intern — May 2025 – Dec. 2025
Toyota Industries Commercial Finance | Python, SQL Server, Azure, Power BI — Dallas, TX
- Built SQL Server ETL and reconciliation workflows integrating PayNet inquiries with internal loan servicing data for recurring credit-risk analytics and reporting.
- Engineered borrower, contract, and asset-level features including LTV ratio, account tenure, inquiry patterns, vehicle segments, and servicing attributes.
- Automated monthly model refreshes with parameterized SQL and Python workflows connected to Power BI dashboards, reducing manual reporting effort by 10%.
- Validated Logistic Regression and Random Forest outputs with ROC/AUC and feature importance, achieving **AUC > 0.82** and improving high-risk identification by **3–4%**.
- Backtested predicted risk buckets against observed outcomes, identifying up to **3.7× higher default rates** in high-risk groups for scorecard validation.

Financial Data Development Engineer – Fraud Detection — Jul. 2024 – Aug. 2024
Caitong Securities Ltd | SQL, Oracle, Stored Procedures — Hangzhou, China
- Designed SQL ETL pipelines processing **10M+ daily transactions** with window functions, subqueries, and stored procedures, improving extraction efficiency by 5%.
- Implemented fraud-detection logic and customer risk scoring from transactional and behavioral streams, reducing false positives by 2% in daily monitoring workflows.
- Built reviewer-facing risk logic for fraud analytics and AML-related workflows, translating transaction patterns into prioritized risk signals for operational review.

Software Engineering Intern – Customer Analytics — Nov. 2023 – Jun. 2024
Joblogic-X Corporation | Python, APIs, Git — Dallas, TX
- Engineered a Python pipeline to scrape and process XHS social-media data, joining unstructured engagement signals with transactional order systems for customer analytics.
- Implemented ranking and clustering logic, integrated KOC scores through REST APIs, and maintained Git workflows for reproducible customer-priority updates.
- Connected ranking outputs to a customer-facing chatbot workflow, translating analytical scores into repeatable customer recommendation data.

Data Scientist Assistant Intern – Image Restoration — Jul. 2023 – Sep. 2023
Chinese Academy of Sciences | TensorFlow, Alibaba Cloud, GFPGAN — Beijing, China
- Deployed image-restoration modules on Alibaba Cloud via SSH, implementing VAE loss, Gaussian filters, and LSGAN functions for reconstruction workflows.
- Built a TensorFlow 2 CNN on 3.4k train/validation/test images, achieved **80% accuracy** across 5 categories, and integrated GFPGAN APIs.

Projects

CodeAssist – Full-Stack AI Feedback System | GitHub — Aug. 2025 – Present
University of Texas at Austin | Faculty Project | Flask, React, PostgreSQL, Docker — Austin, TX
- Developed Flask REST APIs and PostgreSQL models for submissions, AI feedback requests, instructor settings, usage limits, cooldown policies, and role-based access.
- Implemented JSON serialization, backend validation, and Alembic schema changes to keep assignment, submission, and feedback data workflows reproducible.
- Built context filtering so code, uploaded files, tests, results, and outputs are sent to AI providers only when instructor policies allow them.
- Created local stress-test runners with server health checks, structured JSON reports, cleanup controls, and coverage for enrollment, submission, grading, and CRUD workflows.

Researcher & TA – Biological Data Analytics — Jan. 2024 – Dec. 2024
University of Texas at Austin | R, Linux, AWS, Docker — Austin, TX
- Processed **250GB** of RNA-seq data with reproducible R/Linux workflows, applying Salmon, RSEM, Trinity, and samtools for transcript quantification.
- Achieved a **98.7% mapping rate** and identified 48 key genes with Voom/Limma differential-expression analysis and permutation-based FDR.
- Automated normalization, PCA, and volcano-plot workflows in R/ggplot2, improving interpretation efficiency by 15% for time-series biological datasets.

FinChat – AI-Powered Financial Analysis System — Jan. 2023 – May 2024
Southern Methodist University | Python, FinGPT, LangChain, CrewAI — Dallas, TX (Remote Hybrid)
- Built ETL workflows to extract and process 100+ SEC filings from EDGAR, structuring unstructured financial reports for metric extraction and trend detection.
- Implemented LangChain and CrewAI workflows for contextual financial interpretation, improving filing-analysis accuracy by 3% and reducing manual review time.
- Connected filing preprocessing, retrieval, metric extraction, and financial-question workflows into a reusable prototype for comparing companies across reporting periods.